Reasoning About Intuitionistic Alternating-Time Temporal Logic

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Abstract

Multi-Agent Systems (MAS) require formalisms capable of handling partial and evolving information. While imperfect-information extensions of Alternating-time Temporal Logic (ATL) are highly expressive, they typically make model checking undecidable. This paper summarises *Intuitionistic ATL* (IATL), an extension of ATL that replaces classical propositional truth with intuitionistic truth, capturing a novel, computationally tractable notion of imperfect information based on *information refinement.* We summarize IATL's syntax and semantics over *birelational concurrent game structures* (BCGS), the truth-monotonicity characterisation, and the PTIME-completeness of model checking, matching the complexity of classical ATL.

Keywords

Alternating-Time Temporal Logic, Intuitionistic Logic, Multi-Agent Systems, Model Checking

1. Introduction

Formal methods for strategic reasoning play a fundamental role in the specification and verification of Multi-Agent Systems (MAS) [1, 2, 3, 4, 5]. This line of research originated from the use of temporal logics for the specification of reactive systems [6, 7, 8], traditionally interpreted over Kripke structures. The need to reason about open systems with multiple autonomous agents led to the development of strategic formalisms [1, 9, 10, 11, 12], among which Alternating-time Temporal Logic (ATL) [1] stands as a central reference, enabling the specification of cooperative and competitive strategies among agents.

Imperfect information is pervasive in realistic MAS scenarios [13, 14, 15, 16], yet its integration with ATL is notoriously problematic: when agents employ strategies with perfect recall, model checking becomes undecidable [17]. Existing tractable alternatives, such as memoryless or bounded-memory strategies [18, 19, 20, 21, 22, 23, 24, 25, 26, 27], mitigate this at the cost of expressiveness.

Classical propositional logic assumes complete information: the law of the excluded middle $\varphi \vee \neg\varphi$ holds universally, so every formula is either true or false in a model. This assumption is inadequate for agents operating under partial or evolving knowledge [28, 29, 30]. Intuitionistic logic (IL) [31, 32, 33] drops this assumption, making truth procedural and dependent on the ability to verify a formula. The best known semantics for IL is based on Kripke models [34], where the accessibility relation is a partial order modelling progressive information accumulation: a formula is true at a state s only if it remains true in all more informative states reachable from s. This yields a natural *truth-monotonicity* property, making IL a fit base for reasoning about information refinement. Intuitionistic extensions of modal logics have been explored in [35, 36, 37] via birelational Kripke models, and more recently intuitionistic versions of temporal frameworks have been studied for LTL [38, 39] and CTL [40, 41]. Additionally, IL has played a role in non-monotonic reasoning via the Answer Set Programming paradigm [42] and its temporal extensions [43, 44]. The most recent development in this thread is the extension of IL to strategic reasoning in MAS, yielding IATL [45].

This paper summarises *Intuitionistic ATL* (IATL) [45], an extension of ATL that replaces classical propositional truth with intuitionistic truth to obtain a tractable forr of imperfect information. One

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of the core features of IATL is its ability to model *information refinement*: by equipping standard concurrent game structures with a partial order over states, agents can reason about strategies that are winning across all epistemic refinements of their current state, capturing the gradual accumulation of knowledge over time. This is formalised via *birelational concurrent game structures* (BCGS). Unlike imperfect-information ATL, where model checking is undecidable [17], IATL [45] achieves PTIME-completeness, matching classical ATL [1] and making it the first imperfect-information extension of ATL with tractable model checking. We recall IATL's syntax, semantics, and truth-monotonicity characterisation, and discuss an efficient implementation in the VITAMIN model checker [46], with benchmarks confirming practical scalability; we refer to [45] for full details.

IATL exploits intuitionistic truth to represent states where a fact may be unknown, yet sufficient information may still be available to guarantee a winning strategy. Unlike imperfect-information ATL, the law of the excluded middle does not hold in IATL, and model checking remains PTIME-complete. The two frameworks are expressively incomparable: imperfect-information ATL can explicitly model information unavailable to agents, making it suitable for information-flow security analysis, whereas IATL prioritises tractability and the modelling of incremental knowledge acquisition.

2. Preliminaries

We fix a finite set AP of atomic propositions, a finite set Ag of agents, and an action set Act. We assume familiarity with standard ATL [1] and recall only what is needed for IATL.

Intuitionistic Propositional Logic. An *Intuitionistic Kripke Structure* (IKS) is a tuple $\mathcal{K} = \langle S, S_I, \preceq, \mathrm{V}\rangle$, where $\preceq$ is a partial order over S modelling information refinement and $\mathrm{V} : S \to 2^{\mathsf{AP}}$ is monotone w.r.t. $\preceq$. The satisfaction relation for IPL formulas is standard except for implication:

$$\mathcal{K}, s \models \varphi_1 \to \varphi_2 \;\Leftrightarrow\; \forall\, t \geq s \;:\; \mathcal{K}, t \models \varphi_1 \Rightarrow \mathcal{K}, t \models \varphi_2.$$

This yields *truth monotonicity* ($s \preceq t$ and $\mathcal{K}, s \models \varphi$ imply $\mathcal{K}, t \models \varphi$) and refutes the excluded middle $\varphi \vee \neg\varphi$.

Concurrent Game Structures and Strategies. A CGS $\mathcal{G} = \langle S, S_I, \delta, \mathrm{V}\rangle$ is a standard multi-agent game structure [1] with transition function $\delta : S \times \mathrm{Act}^{\mathrm{Ag}} \to S$. Strategies, plays, histories, and outcome function $\mathrm{out}(s, f_A)$ are defined as in [1]; we consider both perfect-recall and memoryless strategies.

3. Intuitionistic ATL

Syntax. Syntax of IATL is as for standard ATL, with the addition of the intuitionistic implication $\to$:

$$\varphi \;::=\; \bot \mid p \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \varphi \to \varphi \mid \langle\!\langle A \rangle\!\rangle \psi \mid [\![A]\!]\, \psi \qquad \psi \;::=\; \bigcirc\varphi \mid \varphi \mathsf{U} \varphi \mid \varphi \mathsf{R} \varphi$$

where $p \in \mathsf{AP}$, $A \subseteq \mathrm{Ag}$, and $\langle\!\langle A \rangle\!\rangle / [\![A]\!]$ are the existential/universal strategic quantifiers. Unlike classical ATL, $\langle\!\langle A \rangle\!\rangle$ and $[\![A]\!]$ are *not* interdefinable in IATL, nor are U and R.

Birelational Concurrent Game Structures. IATL formulas are interpreted over *Birelational CGS* (BCGS), which extend CGS with a partial order $\preceq$ over states modelling information refinement. Formally, a BCGS is a tuple $\mathcal{G} = \langle S, S_I, \delta, \preceq, \mathrm{V}\rangle$ where $\langle S, S_I, \delta\rangle$ is a CGF, $\preceq$ is a partial order over S, and $\mathrm{V} : S \to 2^{\mathsf{AP}}$ satisfies the *monotonicity condition*: $s \preceq t \Rightarrow \mathrm{V}(s) \subseteq \mathrm{V}(t)$.

A BCGS is *well-behaved* if $\preceq$ satisfies two conditions regulating the interplay between information refinement and agent actions. Let $\mathrm{Dec}(s, A)$ denote the A-decisions available at state s:

(C_1) For all $s \preceq s'$ and each $d_A \in \mathrm{Dec}(s, A)$, there exists $d'_A \in \mathrm{Dec}(s', A)$ such that for each full decision d' extending d'_A, there is a full decision d extending d_A with $\delta(s, d) \preceq \delta(s', d')$.

(C_2) The symmetric condition, with $s \preceq s'$ replaced by $s' \preceq s$ and the successor ordering reversed.

Intuitively, C_1 ensures that $\preceq$ behaves as an alternating-time simulation [1]: if s' is more informative than s, then every A-decision at s has a counterpart at s' whose outcomes are at least as informative. C_2 is the dual condition. Well-behavedness can be checked in polynomial time on finite structures [1, 45]. The semantics of IATL extends that of ATL by interpreting implication intuitionistically.

Theorem 1. *Every IATL-satisfiable formula is also ATL-satisfiable, but not vice versa. Adding the law of the excluded middle $\varphi \vee \neg\varphi$ causes IATL to collapse to standard ATL.*

Theorem 2 (Truth Monotonicity)**.** *The key semantic property of IATL is that truth is preserved under information refinement. A BCGS is* IATL-monotonic *if $\mathcal{G}, s \models \varphi$ and $s \preceq t$ imply $\mathcal{G}, t \models \varphi$ for all states s, t and formulas φ.*

The following characterisation shows that well-behavedness is precisely the right structural condition:

Theorem 3 (IATL-monotinicity)**.** *Let $\mathcal{F}$ be a BCGF. Then:* (i) *$\mathcal{F}$ satisfies* C_1 *iff it is $IATL_\exists$-monotonic;* (ii) *$\mathcal{F}$ satisfies* C_2 *iff it is $IATL_\forall$-monotonic;* (iii) *$\mathcal{F}$ is well-behaved iff it is IATL-monotonic.*

Theorem 4. *$IATL_\exists$ and $IATL_\forall$ are expressively incomparable, and $IATL_{\mathsf{U}}$ and $IATL_{\mathsf{R}}$ are expressively incomparable, independently of the well-behaved assumption.*

Theorem 5. *For any finitely-branching BCGS $\mathcal{G}$, state s of $\mathcal{G}$, and IATL formula φ: $\mathcal{G}, s \models \varphi \Leftrightarrow \mathcal{G}, s \models_m \varphi$, independently of the well-behaved assumption.*

For a formula φ and a BCGS $\mathcal{G}$, we denote by $[\![\varphi]\!]_{\mathcal{G}}$ the set of states satisfying φ. Given $X \subseteq S$, we write $X^{\mathbf{c}}$ for its complement and $X^{\uparrow}$ for the upward closure $\{s \in S \mid \forall s' \geq s,\ s' \in X\}$. Moreover, for a coalition A, we write $\mathrm{Pre}^{\exists}(A, X)$ for the set of states s such that for *some* A-decision d_A available at s, *all the successors* of s consistent with d_A are in X. Finally, we denote by $\mathrm{Pre}^{\forall}(A, X)$, the set of states s such that for *all* A-decisions d_A available at s, *some successor* of s consistent with d_A is in X. The following proposition easily follows from the semantics of IATL.

Proposition 1. *Given a BCGS $\mathcal{G}$ with set of states S and a coalition A, the following holds:*
- $[\![\varphi_1 \rightarrow \varphi_2]\!] = ([\![\varphi_1]\!]^{\mathbf{c}} \cup [\![\varphi_2]\!])^{\uparrow}$.
- $[\![\langle\!\langle A \rangle\!\rangle \bigcirc \varphi]\!] = Pre^{\exists}(A, [\![\varphi]\!])$.
- $[\![[\![A]\!] \bigcirc \varphi]\!] = Pre^{\forall}(A, [\![\varphi]\!])$.
- $(Pre^{\forall}(A, [\![X]\!]))^{\mathbf{c}} = Pre^{\exists}(A, [\![X]\!]^{\mathbf{c}})$ *for each $X \subseteq S$.*

The above characterisations are specific to IATL; denotational and fixpoint identities are as in ATL [1].

4. IATL Model Checking

The *model-checking problem* for IATL asks, given a finite BCGS $\mathcal{G}$ and an IATL formula Φ, whether $\mathcal{G}, s \models \Phi$ for each initial state s.

Theorem 6 (IATL Model Checking Complexity)**.** *IATL model checking is* PTIME*-complete and solvable in time $O(|\mathcal{G}|^2 \cdot |\Phi|)$.*

PTIME-hardness follows from PTIME-hardness of ATL model checking [1]: any CGS $\mathcal{G}$ can be viewed as a BCGS equipped with the identity partial order, reducing ATL model checking to IATL model checking. For the upper bound, the algorithm computes the denotation $[\![\varphi]\!]$ of each subformula φ of Φ by a bottom-up traversal of its parse tree. All cases coincide with standard ATL model checking, except for intuitionistic implication, handled via upward closure:

$$[\![\varphi_1 \rightarrow \varphi_2]\!] \ := \ ([\![\varphi_1]\!]^{\mathsf{C}} \cup [\![\varphi_2]\!])^{\uparrow}.$$

Each subformula is processed once; computing the upward closure $X^{\uparrow}$ takes $O(|S|^2)$, yielding the overall $O(|\mathcal{G}|^2 \cdot |\Phi|)$ bound.

Implementation. The algorithm has been implemented by extending the VITAMIN model checker [46] to support IATL semantics. Benchmarks on the muddy-children scenario [47] confirm practical scalability as the number of agents grows; reference to [45] for full experimental details.

5. Conclusion and Future Work

This paper summarises IATL [45], an extension of ATL that replaces classical propositional truth with intuitionistic truth to model information refinement in MAS. Unlike classical ATL, which assumes complete knowledge, IATL allows facts to be neither true nor false at a state, yet still guarantees the existence of winning strategies across all epistemic refinements. This makes IATL suitable for realistic scenarios such as sensing, incremental knowledge acquisition, and distributed protocol verification. Crucially, model checking remains PTIME-complete, matching classical ATL while gaining expressive power for imperfect-information settings. More broadly, IATL sits at the intersection of formal methods and artificial intelligence: by providing a tractable logical foundation for reasoning under partial knowledge, it contributes to the growing agenda of formal approaches to AI verification [2, 3], particularly in safety-critical MAS applications where incomplete information is the rule rather than the exception. Several directions are open. A complete axiomatization of IATL would unlock satisfiability and validity checking. Extending the intuitionistic approach to more expressive logics such as ATL* or Strategy Logic [3] is another natural avenue. A promising direction is the integration of IATL with quantitative frameworks. In particular, combining intuitionistic truth with the fuzzy semantics and resource-bounded natural strategies of HumanATL[$\mathscr{F}$] [48] would yield a framework where strategic truth is simultaneously graded, capturing partial achievement of goals, and monotone with respect to information refinement, offering a unified treatment of epistemic and quantitative uncertainty in human-like MAS reasoning. Another avenue concerns the application of IATL to dynamic, real-world environments. Recent work [49] has demonstrated the feasibility of deploying ATL model checking directly within game engines for real-time NPC decision-making. In such settings, agents typically operate under asymmetric information, a scenario that classical ATL cannot handle but that falls naturally within the scope of IATL. Extending the in-engine framework of [49] with IATL semantics would allow NPCs to reason strategically under genuine imperfect information, opening the door to more realistic and formally grounded game AI. The neurosymbolic architecture proposed in [50], which integrates an autoencoder with Obstruction Logic for explainable medical diagnosis, offers a further connection. The diagnostic process modelled there, where a logical component progressively eliminates hypotheses incompatible with observed evidence, is conceptually aligned with the information refinement paradigm of IATL. A natural question is whether the neurosymbolic framework of [50] could be instantiated with IATL as the symbolic reasoning component, replacing hypothesis elimination with epistemic refinement, ensuring that diagnostic strategies remain robust not only with respect to current evidence, but also to any future observations that may refine the clinical picture. Taken together, these directions point toward a broader vision in which IATL serves as a foundational building block for the next generation of formal AI systems: tractable, epistemically aware, and compositionally integrable with quantitative, neurosymbolic, and real-time reasoning frameworks. Finally, IATL could be paired with *neuro-fuzzy* architectures such as [51], whose graded truth values would feed a quantitative, fuzzy variant of the logic in the spirit of quantitative ATL [52]. Here the neural component acts as a sub-symbolic front-end emitting graded evidence, while fuzzy IATL serves as a symbolic layer combining monotonicity under information refinement with graded satisfaction of strategic goals, unifying epistemic and quantitative uncertainty in a single learnable framework.

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